

- 1** One end of a light elastic string, of natural length a and modulus of elasticity $3mg$, is attached to a fixed point O . The other end of the string is attached to a particle P of mass m . The string hangs with P vertically below O . The particle P is pulled vertically downwards so that the extension of the string is $2a$. The particle P is then released from rest.

- (a)** Find the speed of P when it is at a distance $\frac{3}{4}a$ below O . [3]

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- (b)** Find the initial acceleration of P when it is released from rest. [2]

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